

Automation Design with Salesforce



Salesforce CRM Platform Implementation for Customer Operations and Service Management GreenGrid Utilities

6.1 Flow Design

To address the operational inefficiencies identified during the business analysis conducted in Milestone 2, particularly those related to technician dispatch coordination and delayed service status synchronization, GreenGrid Utilities adopts a declarative automation strategy using **Salesforce Record-Triggered Flows**.

The objective of this automation framework is to reduce manual intervention, improve process consistency, and establish seamless communication between customer support personnel and field service technicians. Through the implementation of automated workflows, the organization can accelerate service response times, improve data accuracy, and provide real-time operational visibility across departments.

The automation framework consists of two primary flows that support the end-to-end service management lifecycle.

6.1.1 Flow 1: Automated Field Dispatch Process (Dispatch Engine)

- **Triggering Object:** Case
- **Trigger Source Event:** A record is created or updated.
- **Trigger Condition Criteria:** Priority equals High, Status equals New, and Utility_Asset__c contains a valid lookup value.
- **Optimization Goal:** Rapid Service Dispatch Processing (Actions & Related Records).
- **Architectural Blueprint:**

When a customer support representative receives a critical service request and creates a high-priority Case associated with an existing Utility Asset, Salesforce automatically evaluates the record against the defined trigger criteria. Once the conditions are satisfied, the automation process creates a related Field_Visit__c record without requiring manual intervention.

The newly created Field Visit record is automatically linked to the originating Case through the Parent_Case__c relationship field. Essential service information is transferred to the field service process, allowing technicians to access relevant customer and asset information immediately.

This automation eliminates delays caused by manual dispatch procedures and ensures that service requests are routed to field operations as quickly as possible. The result is improved service responsiveness, reduced administrative workload, and enhanced coordination between support and technical teams.

6.1.2 Flow 2: Service Resolution Synchronization and Automatic Case Closure (Sync Engine)

- **Triggering Object:** Field Visit (Field_Visit__c)
- **Trigger Source Event:** A record is updated.
- **Trigger Condition Criteria:** Repair_Status__c equals Resolved and the status value has changed.
- **Optimization Goal:** Real-Time Service Status Synchronization (Actions & Related Records).
- **Architectural Blueprint:**

After completing repair or maintenance activities, field technicians update the corresponding Field Visit record using Salesforce Mobile. When the Repair Status field is changed to "Resolved", Salesforce immediately activates the synchronization flow.

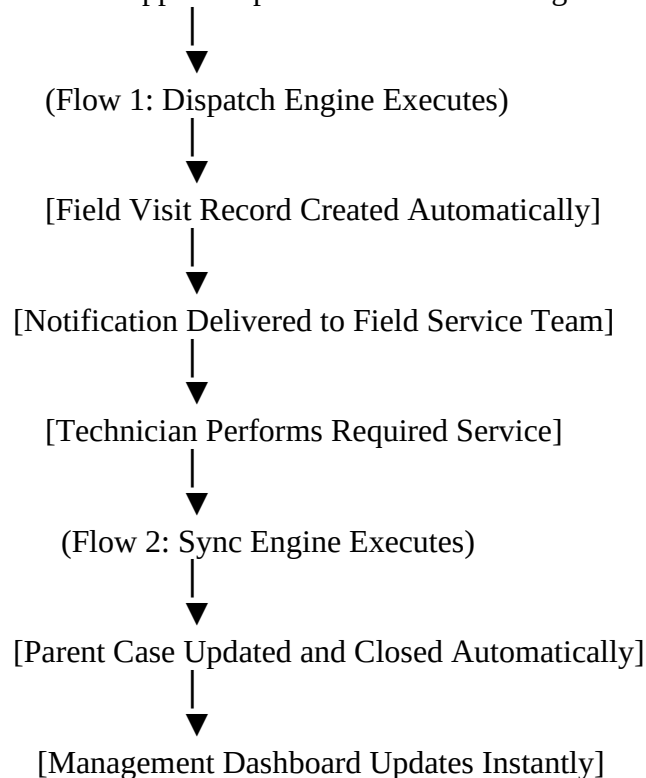
The automation retrieves the related parent Case and automatically updates its status to Closed. Technician observations and service completion details can also be transferred from the Field Visit record to the Case record to maintain a complete service history.

In addition, the system records service completion information that can be used for reporting, dashboard visualization, and operational performance monitoring. By automating this process, GreenGrid Utilities eliminates reporting delays and ensures that customer service records always reflect the latest field activity information.

6.2 Workflow Documentation

6.2.1 Core Functional Workflows

[Customer Support Representative Creates High-Priority Case]



a. Service Request Intake and Automated Dispatch (Flow 1 Workflow)

1. User Action

A Customer Support Representative receives a customer request regarding a service disruption or operational issue. The representative identifies the relevant Utility Asset using the associated Meter ID and creates a new Case record with Priority set to High and Status set to New.

2. Automated Evaluation

Immediately after the record is saved, Salesforce evaluates the Case information against the criteria configured within Flow 1. The system verifies that the Case satisfies all required conditions before initiating the automation process.

3. Record Creation

Once validation is completed successfully, Salesforce automatically generates a new Field_Visit__c record. The flow establishes the relationship between the Case and Field Visit records by assigning the Case identifier to the Parent_Case__c field.

The automation also initializes service-related values such as Repair_Status__c and prepares the record for technician assignment.

4. Notification Distribution

After the Field Visit record has been created, the system can distribute notifications to the field service team, ensuring that technicians are informed of new assignments and can begin service activities without delay.

b. Field Service Completion and Automated Case Closure (Flow 2 Workflow)

1. User Action

After arriving at the service location and completing the required repair work, the Field Technician accesses the assigned Field Visit record through Salesforce Mobile. The technician enters the necessary service notes and updates the Repair Status field to Resolved.

2. Automated Evaluation

When the updated record is saved, Salesforce automatically evaluates the change and determines whether the conditions for Flow 2 have been satisfied.

3. Parent Record Synchronization

The automation locates the related Case record through the Parent_Case__c lookup relationship and performs an immediate update. The Case status is changed to Closed and technician comments may be transferred to the appropriate Case field.

4. Operational Metrics Update

The system records service completion information and updates operational reporting data. This information supports performance measurement, management reporting, and executive dashboard monitoring activities.

6.3 Error Handling Strategy

To support reliable system performance and minimize operational disruptions, GreenGrid Utilities incorporates structured error-handling mechanisms into each automation process. These controls help ensure that automation failures are detected, reported, and addressed quickly.

6.3.1 Fault Path Management

Every critical Flow element responsible for database operations, including Create Records and Update Records actions, incorporates a dedicated Fault Path configuration.

If Salesforce encounters a processing issue such as a validation rule failure, record locking conflict, missing data dependency, or system timeout, the Fault Path intercepts the exception and redirects execution to a predefined error-handling process.

This approach prevents unexpected flow termination and improves overall automation reliability.

6.3.2 Automated Error Notification and Logging

Whenever a Fault Path is activated, Salesforce executes predefined notification and logging procedures to assist with troubleshooting and issue resolution.

1. Administrator Email Notification

A Send Email action automatically delivers an alert message to the Salesforce System Administrator.

The notification includes:

- Flow Name
- Record Identifier
- User Information
- Execution Timestamp
- Detailed Error Message
- Flow Fault Description

This information enables administrators to identify and resolve issues efficiently.

2. Internal Monitoring Notification

In addition to email alerts, automation errors can be logged within internal monitoring channels to ensure visibility across administrative teams.

This additional layer of monitoring supports incident tracking and helps ensure that critical automation failures receive prompt attention.

6.4 Salesforce Implementation and Configuration Steps

The following section outlines the detailed procedures required to configure, deploy, test, and activate the automation framework within the Salesforce Developer Edition environment. These implementation activities focus on the creation of two Record-Triggered Flows that support customer service operations and field service management processes at GreenGrid Utilities.

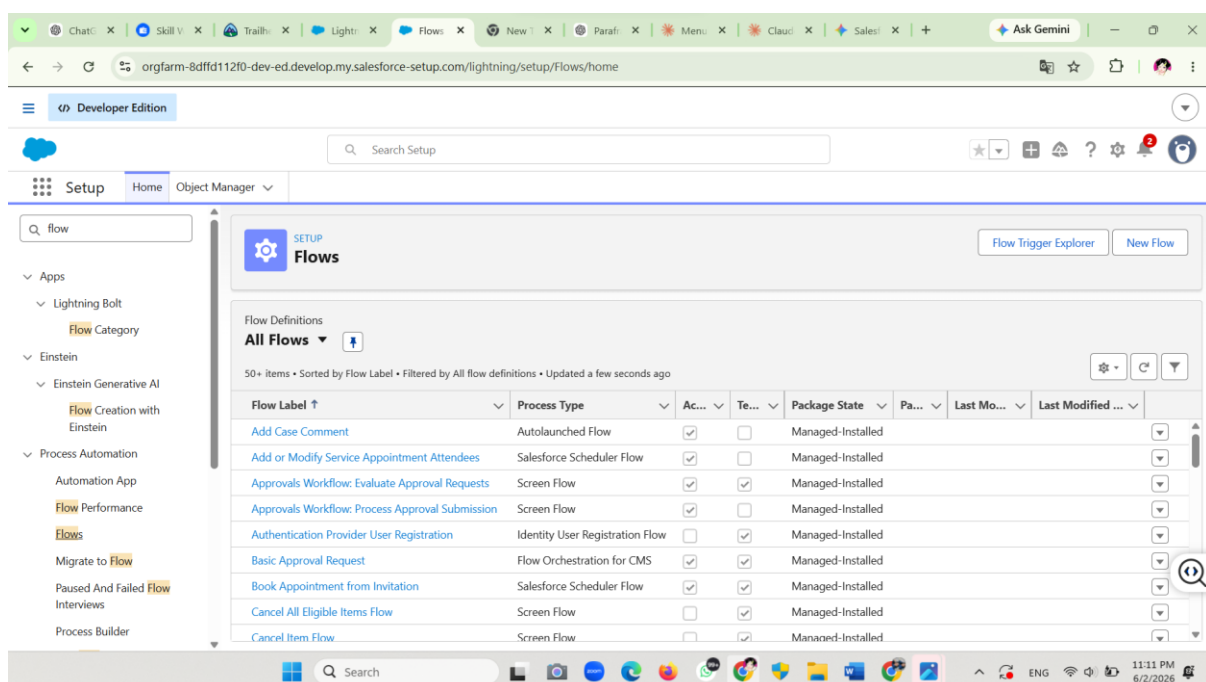
Step 1: Access Salesforce Flow Builder

Purpose

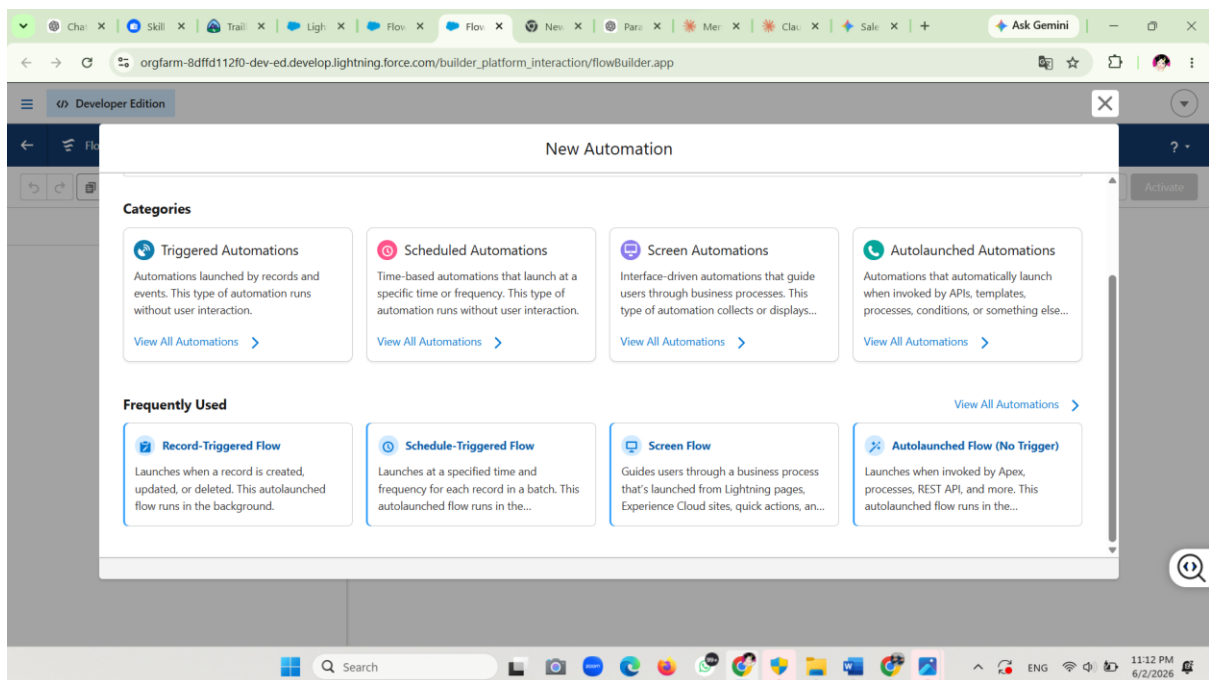
The purpose of this step is to access Salesforce Flow Builder, which serves as the primary platform tool for creating and managing automation processes.

Procedure

1. Log in to the Salesforce Developer Edition account.
2. Click the **Setup** gear icon located in the upper-right corner of the screen.
3. Select **Setup** from the menu.
4. In the **Quick Find** search box, type **Flows**.



5. Click **Flows** under the **Process Automation** section.



Step 2: Create Flow 1 – Automated Field Dispatch Process

Purpose

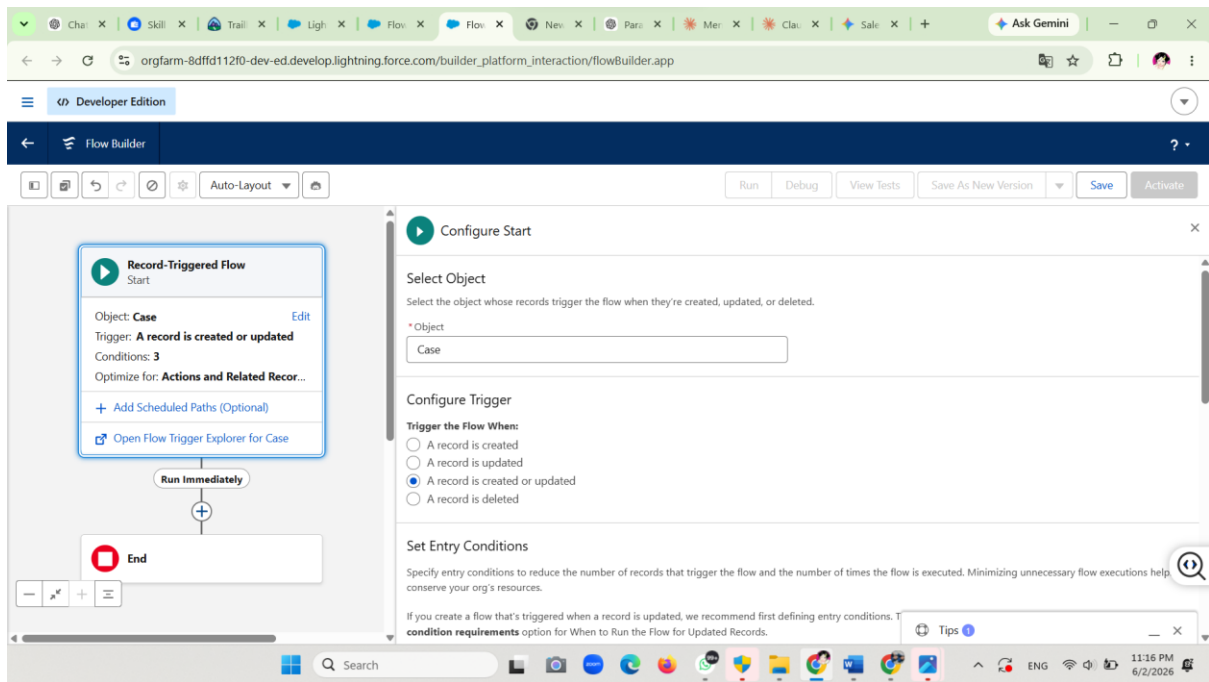
This automation flow is designed to automatically create a Field Visit record whenever a high-priority customer service Case is submitted.

Procedure

1. Click **New Flow**.
2. Select **Record-Triggered Flow**.
3. Click **Create**.

Configure the Start element using the following settings:

Configuration	Value
Object	Case
Trigger	A Record Is Created or Updated
Optimize For	Actions and Related Records



Configure Entry Conditions

Select:

All Conditions Are Met (AND)

Configure the conditions below:

Field	Operator	Value
Priority	Equals	High
Status	Equals	New
Utility_Asset__c	Is Null	False

Configure Execution Behavior

Under:

When to Run the Flow for Updated Records

Select:

Only when a record is updated to meet the condition requirements

Step 3: Add Create Records Element

Purpose

The purpose of this element is to automatically generate a related Field Visit record.

Procedure

1. Click the + icon below the Start element.
2. Select **Create Records**.

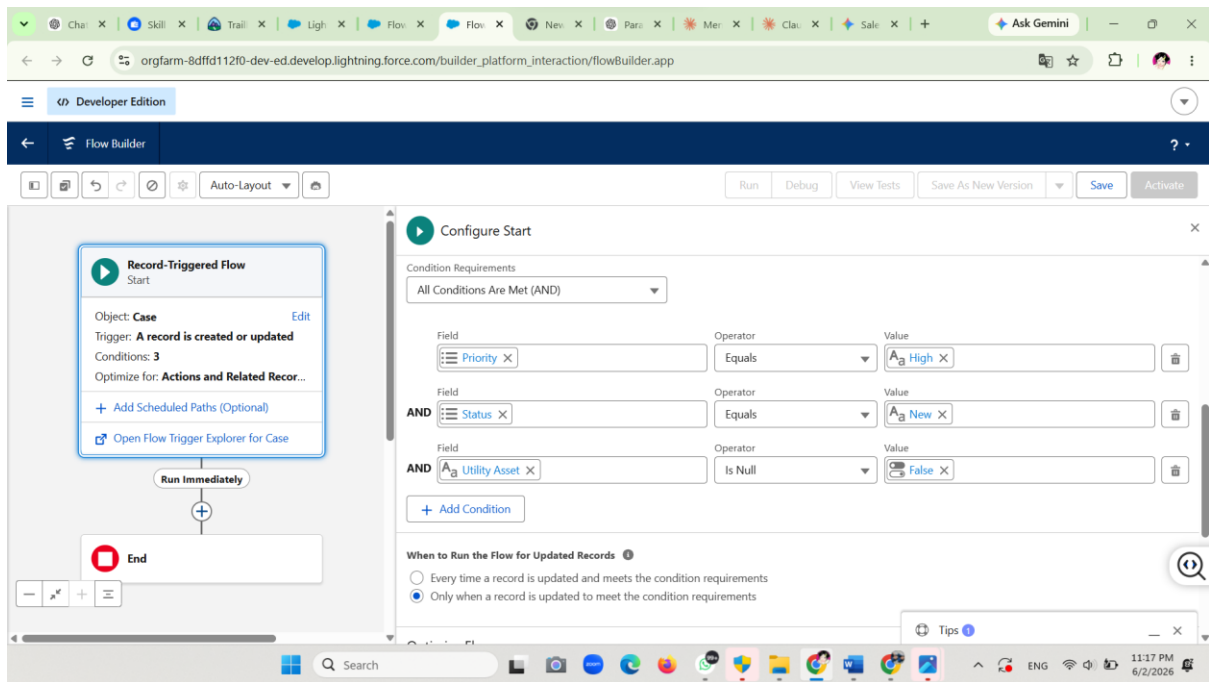
Configure the element as follows:

Setting	Value
Label	Create Related Field Visit
API Name	Create_Related_Field_Visit
Create Records	One Record
Object	Field_Visit__c

Configure Field Mapping

Field Visit Field	Value
Parent_Case__c	{!\$Record.Id}
Repair_Status__c	Assigned

The screenshot displays the Salesforce Flow Builder interface. On the left, a flow diagram shows a 'Record-Triggered Flow' starting with a 'Run Immediately' trigger, followed by a 'Create Related Field Visit' element (highlighted with a blue border), and ending with an 'End' node. On the right, the configuration panel for the 'Create Records' element is open. It includes fields for 'Label' (Create Related Field Visit) and 'API Name' (Create_Related_Field_Visit). The 'Description' field is empty. Under 'How to set record field values', the 'Manually' option is selected. The 'Create a Record of This Object' section shows 'Field_Visit' as the object. The 'Set Field Values for the Field_Visit' section shows a mapping for 'Parent Case' with a value of 'A Tr'.



Step 4: Configure Fault Path for Flow 1

Purpose

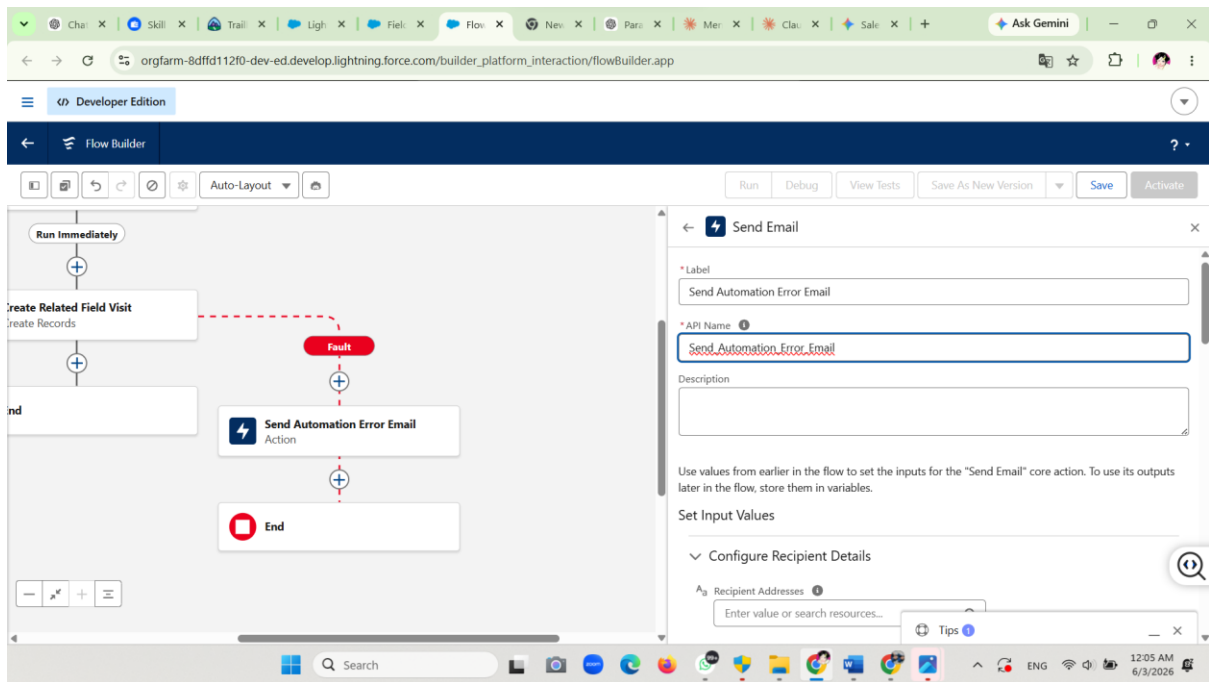
This configuration ensures that any automation failures are properly captured and reported to system administrators.

Procedure

1. Select the **Create Related Field Visit** element.
2. Click **Add Fault Path**.
3. On the Fault Path branch, click +.
4. Select **Action**.
5. Search for **Send Email**.

Configure the action using the following settings:

Setting	Value
Label	Send Automation Error Email
API Name	Send_Automation_Error_Email
Subject	GreenGrid Utilities Automation Failure
Recipient	System Administrator Email
Body	Include {\$Flow.FaultMessage}



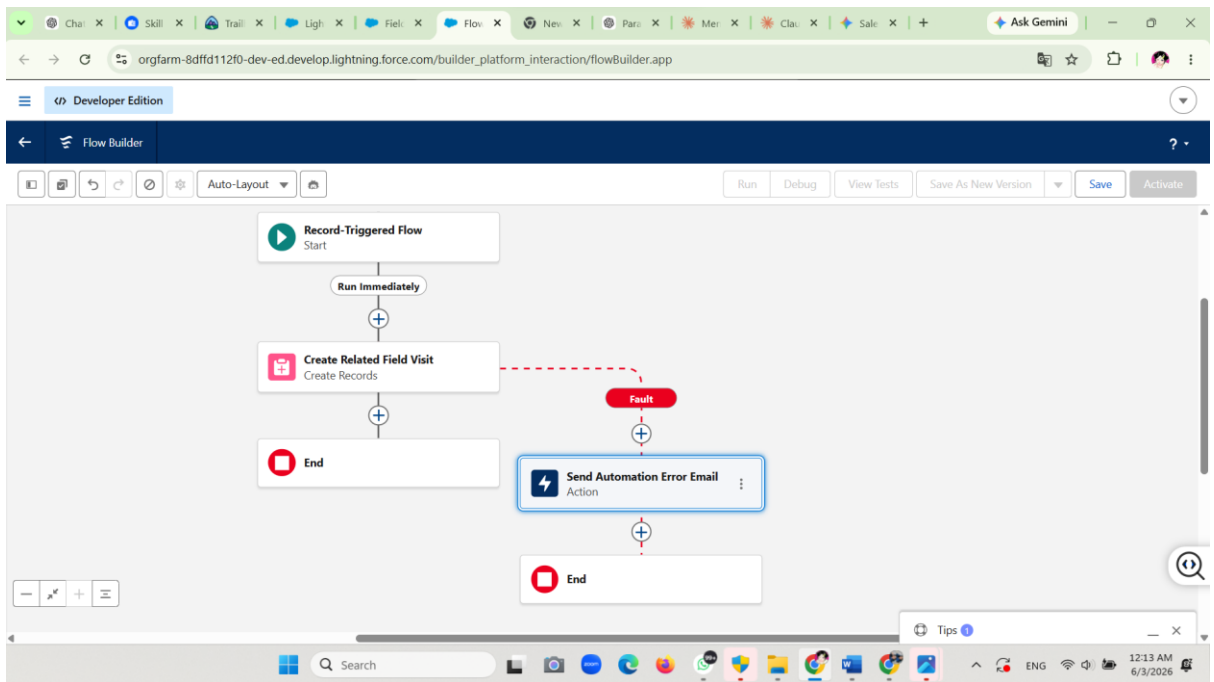
Step 5: Save and Activate Flow 1

Procedure

1. Click **Save**.
2. Enter the following Flow Label:

Case Automated Field Dispatch Trigger

3. Verify the generated API Name.
4. Click **Save**.
5. Click **Activate**.



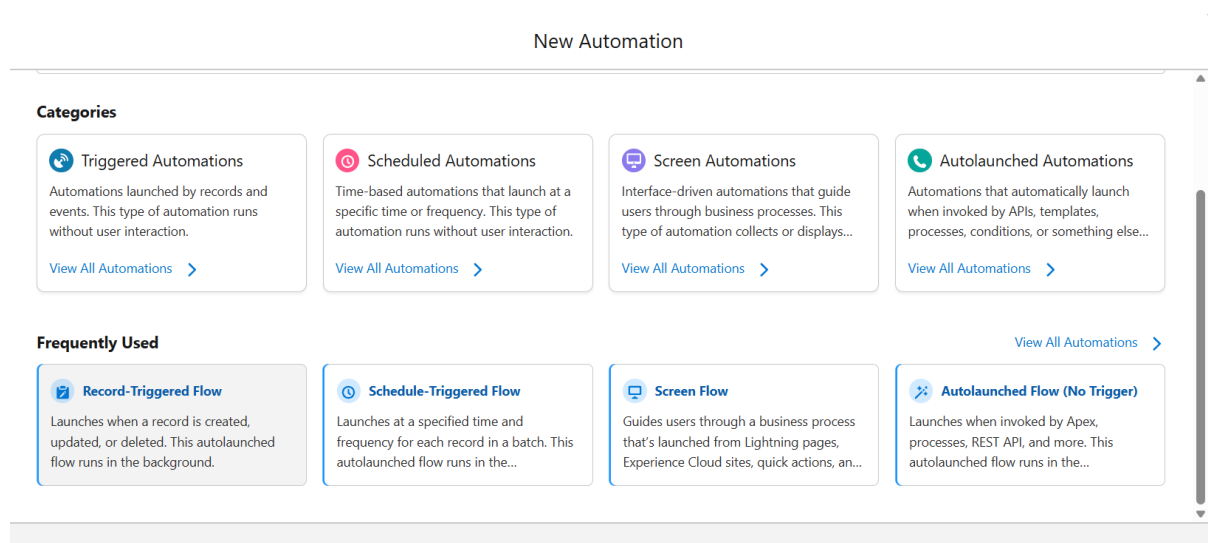
Step 6: Create Flow 2 – Service Resolution Synchronization

Purpose

This automation flow synchronizes completed field service activities with the related customer service Case record.

Procedure

1. Click **New Flow**.
2. Select **Record-Triggered Flow**.
3. Click **Create**.



Configure the Start element as follows:

Configuration	Value
Object	Field_Visit__c
Trigger	A Record Is Updated
Optimize For	Actions and Related Records

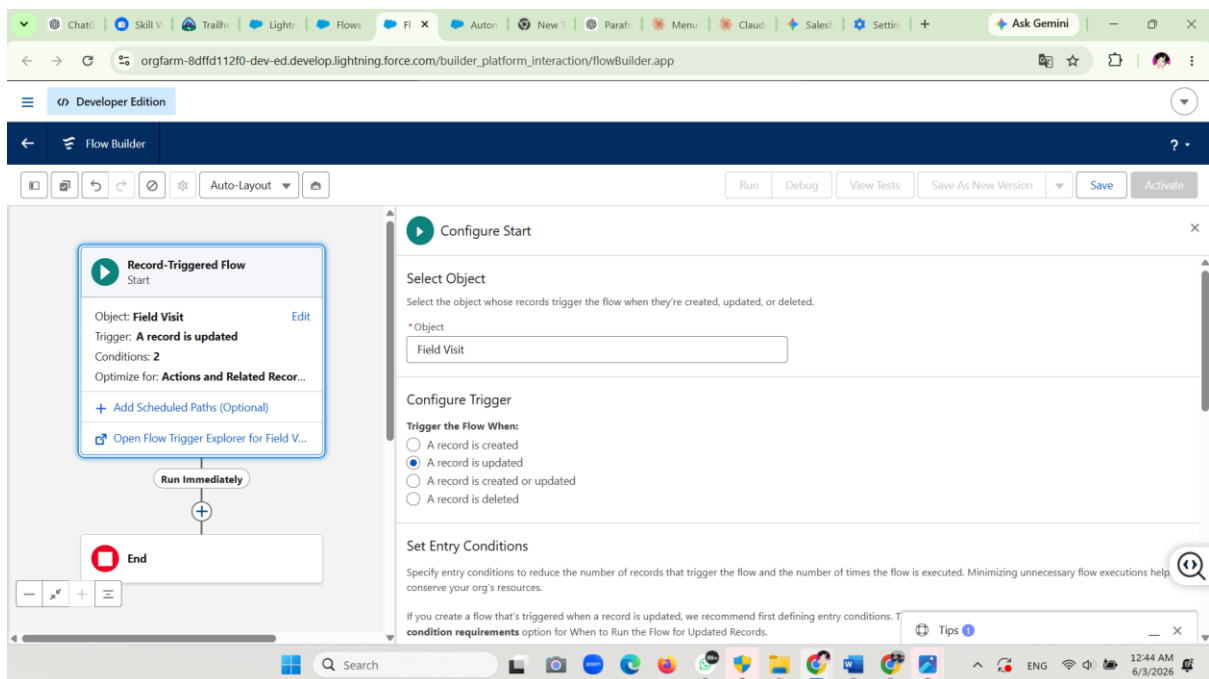
Configure Entry Conditions

Select:

All Conditions Are Met (AND)

Configure:

Field	Operator	Value
Repair_Status__c	Equals	Resolved
Parent_Case__c	Is Null	False



Step 7: Add Update Records Element

Purpose

The Update Records element automatically updates the parent Case when field work has been completed.

Procedure

1. Click the + icon below the Start element.
2. Select **Update Records**.

Configure:

Setting	Value
Label	Update Parent Case to Closed
API Name	Update_Parent_Case_to_Closed

Define Record Criteria

Field	Operator	Value
Id	Equals	{!\$Record.Parent_Case__r.Id}

Configure Field Updates

Case Field	Value
Status	Closed
Description	{!\$Record.Technician_Notes__c}

Step 8: Configure Fault Path for Flow 2

Purpose

This configuration provides error notifications whenever synchronization failures occur.

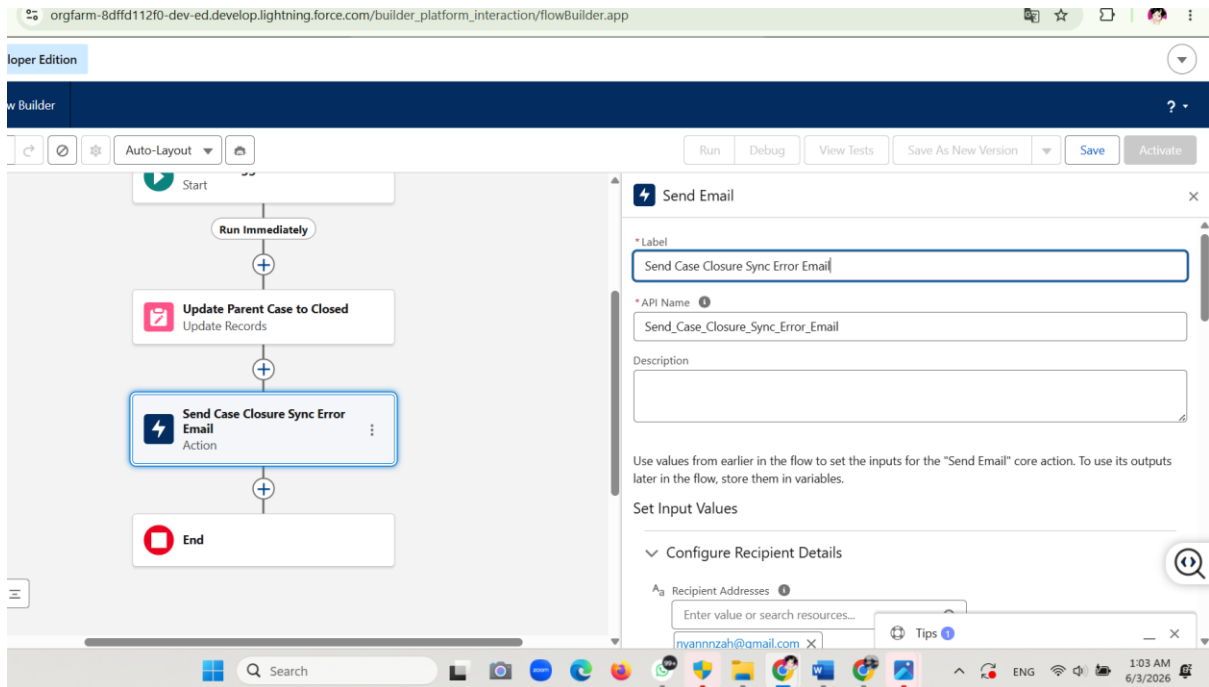
Procedure

1. Select the **Update Parent Case to Closed** element.
2. Add a **Fault Path**.
3. Add a **Send Email Action**.

Configure:

Setting	Value
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Label	Send Case Closure Error Email
API Name	Send_Case_Closure_Error_Email
Subject	GreenGrid Utilities Synchronization Failure
Recipient	System Administrator



Step 9: Save and Activate Flow 2

Procedure

1. Click **Save**.
2. Enter the following Flow Label:

Field Visit Real-Time Operational Sync and Case Closure

3. Click **Save**.
4. Click **Activate**.

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Developer Edition

Flow Builder | Real-Time Operational Sync and Case Closure - V1

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```
graph TD; Start([Record-Triggered Flow  
Start]) --> Trigger([Run Immediately]); Trigger --> Join1((+)); Join1 --> Update[Update Parent Case to Closed  
Update Records]; Update --> Join2((+)); Join2 --> End1([End]); Update -. Fault .-> Fault([Fault]); Fault --> Join3((+)); Join3 --> SendEmail[Send Case Closure Sync Error Email  
Action]; SendEmail --> Join4((+)); Join4 --> End2([End]);
```

Flow Builder diagram showing a Record-Triggered Flow. The flow starts with a 'Record-Triggered Flow' event, followed by a 'Run Immediately' trigger. The main path goes through an 'Update Parent Case to Closed' action (highlighted with a blue border) to an 'End' node. A dashed red line indicates a fault path from the 'Update Parent Case to Closed' action to a 'Fault' node, which then leads to a 'Send Case Closure Sync Error Email' action and finally to another 'End' node.